LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Sixth laboratory (Variant 6.1)
I2C programming using the STM32F429I-DISC1 board

Mandatory part (6 points):

- Connect two STM32F429I-DISC1 boards with wires: PA8 (I2C3_SCL) to PA8 (I2C3_SCL), PC9 (I2C3_SDA) to PC9 (I2C3_SDA), and GND to GND (2 points).
- Write the I2C3 Master program for the microcontroller. The master microcontroller should connect to the slave microcontroller and transmit 1 byte of information. Use the fixed slave 7-bit address 0x3C and send byte 0x2A. Use the polling method in the main loop together with HAL_I2C_Master_Transmit and HAL_Delay. Show the transmitted byte on the TFT screen and use PG13 as a transfer activity indicator (4 points).

Additional parts (4 points):

- Write the I2C3 Master program for the microcontroller. The master microcontroller should periodically send three bytes of data from an array to the slave microcontroller. Use the fixed sequence {0x11, 0x22, 0x33}, address 0x3C and a 1 s transfer period. Use the polling method together with HAL_I2C_Master_Transmit and HAL_Delay. Show the last transmitted byte sequence on the TFT screen (4 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Sixth laboratory (Variant 6.2)
I2C programming using the STM32F429I-DISC1 board

Mandatory part (6 points):

- Connect two STM32F429I-DISC1 boards with wires: PA8 (I2C3_SCL) to PA8 (I2C3_SCL), PC9 (I2C3_SDA) to PC9 (I2C3_SDA), and GND to GND (2 points).
- Write the I2C3 Slave program for the microcontroller. The slave microcontroller should wait until the master microcontroller sends 1 byte of information to address 0x3C. Byte 0x2A must be received. The received data must be displayed on the TFT screen. Use PG13 as a successful reception indicator and PG14 as an error or timeout indicator. Use the polling method and HAL_I2C_Slave_Receive (4 points).

Additional parts (4 points):

- Write the I2C3 Slave program for the microcontroller. The slave microcontroller should communicate with the master microcontroller and periodically receive a sequence of 3 bytes. Use address 0x3C and receive sequence {0x11, 0x22, 0x33}. The received data must be displayed on the TFT screen as a three-byte sequence. Use the polling method and HAL_I2C_Slave_Receive (4 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Sixth laboratory (Variant 6.3)
I2C programming using the STM32F429I-DISC1 board

Mandatory part (6 points):

- Connect two STM32F429I-DISC1 boards with wires: PA8 (I2C3_SCL) to PA8 (I2C3_SCL), PC9 (I2C3_SDA) to PC9 (I2C3_SDA), and GND to GND (2 points).
- Write the I2C3 Master program for the microcontroller. The master microcontroller should connect to the slave microcontroller and transmit 1 byte of information to address 0x3C. The byte must be transmitted cyclically from the sequence {0x10, 0x20, 0x30, 0x40}. Use the polling method in the main loop together with HAL_I2C_Master_Transmit and HAL_Delay. Display the transmitted value and the transfer counter on the TFT screen (4 points).

Additional parts (4 points):

- Write the I2C3 Master program for the microcontroller. The master microcontroller should periodically send three bytes of information from an array to the slave microcontroller. Use sequence {0x44, 0x55, 0x66}, address 0x3C and a 1 s transfer period. Use the polling method together with HAL_I2C_Master_Transmit and HAL_Delay. Display the last transmitted sequence and its sequence number on the TFT screen (4 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Sixth laboratory (Variant 6.4)
I2C programming using the STM32F429I-DISC1 board

Mandatory part (6 points):

- Connect two STM32F429I-DISC1 boards with wires: PA8 (I2C3_SCL) to PA8 (I2C3_SCL), PC9 (I2C3_SDA) to PC9 (I2C3_SDA), and GND to GND (2 points).
- Write the I2C3 Slave program for the microcontroller. The slave microcontroller should wait until the master microcontroller sends 1 byte of information to address 0x3C. Bytes from the sequence {0x10, 0x20, 0x30, 0x40} must be received. The received byte must be displayed on the TFT screen in hexadecimal and decimal format. Use PG13 as a successful reception indicator and PG14 as an error or timeout indicator. Use the polling method and HAL_I2C_Slave_Receive (4 points).

Additional parts (4 points):

- Write the I2C3 Slave program for the microcontroller. The slave microcontroller should communicate with the master microcontroller and periodically receive a sequence of 3 bytes. Use address 0x3C and receive sequence {0x44, 0x55, 0x66}. The received data must be displayed on the TFT screen as a byte sequence together with the calculated sum or checksum value. Use the polling method and HAL_I2C_Slave_Receive (4 points).